



QUANTITATIVE METHODS 1 · CHAPTER 5

Time Series

Exercise series

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1. Components and models

For each description identify trend, seasonality, cycle or irregular component: long-run e-commerce growth; December sales peaks; recession-related movements; a one-day system failure. When is a multiplicative model preferable?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Moving averages

Monthly demand for five months is 90, 105, 99, 120, 126. Calculate the three-period centred moving averages for months 2, 3 and 4.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Linear trend

For $t=1,2,3,4$ and $Y=(52,57,61,66)$ estimate $Y_t=a+bt$ using least squares and forecast $t=5$.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. Multiplicative seasonal forecast

Quarterly trend is $T_t = 100 + 3t$. Seasonal factors are .80, 1.05, .95 and 1.20 for Q1-Q4.

- 1 Check normalisation.
- 2 Forecast $t=9,10,11,12$.
- 3 Interpret the Q4 factor.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. Forecast evaluation

Actual values are 100, 120, 110, 130. Model A forecasts 98, 118, 115, 128; Model B forecasts 105, 112, 108, 140. Calculate MAE for both and recommend a model. What else should be checked?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Centred moving average and seasonal reading

Quarterly sales are 80, 100, 90, 130, 88, 110, 99 and 143.

- 1 Calculate the four-term moving averages.
- 2 Centre the moving averages for quarters 3 to 6.
- 3 Compute the observed/centred-moving-average ratios for quarters 3 to 6.
- 4 Explain what these ratios suggest.

Method prompt. A four-term average lies between quarters; centre it by averaging two consecutive four-term averages.

7. Forecast validation and structural break

Actual monthly demand is 100, 104, 108, 150, 154 and 158. A trend model fitted on the first three months predicts 100, 104, 108, 112, 116 and 120.

- 1 Calculate forecast errors for months 4-6.
- 2 Calculate MAE and mean error over the validation period.
- 3 Diagnose the model.
- 4 Propose a responsible forecasting response.

Method prompt. Use error = actual – forecast. A persistently positive error signals systematic underforecasting.